

RE-MASTERS (RM.Sc.) IN COMPUTER SCIENCE

HWP LABS, BENIN CITY, EDO STATE

EVALUATION EXAMINATION 2018/2019 SESSION

Programme: Slingshot JS

Batch Code: HWP/RMSC/AUG/2019

Exam Schedule: Saturday, March 21, 2020 at 12 Noon

Time Allocated: 2 hours

Student ID: HWP/RMSC/_ _ _

INSTRUCTION: ATTEMPT ALL QUESTIONS

Scope I: Introduction to JavaScript

- JavaScript was developed by _____, in the year _____?
 - Anders Hejlsberg, 2012
 - Brendan Eich, 1995
 - Jeremy Ashkenas, 2011
 - Ryan Dahl, 2009
- JavaScript is a _____ programming language?
 - Functional
 - Logic
 - Procedural
 - Scripting
- What is JavaScript used for _____?
 - Webpage Behavior
 - Webpage Content
 - Webpage Design
 - Webpage Structure
- JavaScript primarily interacts with the _____ layer in a Website Architecture?
 - Controller
 - Model
 - View
 - None of the above
- The _____ tag is used to begin a JavaScript shell?
 - <doctype>
 - <html>
 - <javascript>
 - <script>

Scope II: Standard I/O and Expressions

6. Which of the following “signtax” is used to insert a Comment in JavaScript?
 - A. <!-- TODO documentation -->
 - B. /* TODO documentation */
 - C. // TODO documentation
 - D. # TODO documentation
7. A typical JavaScript *expression* is made up of _____?
 - A. data types and operators
 - B. identifiers and operators
 - C. operands and operators
 - D. variables and operators
8. Identifiers that store values that cannot be changed are declared as _____.
 - A. const
 - B. let
 - C. strict
 - D. var
9. Which of the following keywords defines a variable with a Global scope?
 - A. const
 - B. let
 - C. strict
 - D. var
10. Which of the following keywords defines a variable with a Local scope?
 - A. const
 - B. let
 - C. strict
 - D. var
11. The _____ popup is used to request user input?
 - A. alert()
 - B. confirm()
 - C. console.log()
 - D. prompt()
12. The _____ popup is used to display output?
 - A. alert()
 - B. confirm()
 - C. console.log()
 - D. prompt()

Scope III: Operators

13. Ternary operator ($x ? y : z$) is a form of _____?
- A. Arithmetic operators
 - B. Assignment operators
 - C. Comparison operators
 - D. Logical operators
14. Write a JavaScript program to compute Simple Interest given the following parameters; Principal N240,000, Rate 15% and Tenure 3 years.
- A. N 36,000
 - B. N 108,000
 - C. N 276,000
 - D. N 348,000
15. Write a JavaScript program to compute the roots of a Quadratic Equation given the following parameters; Coefficient A = 1, Coefficient B = 10, Constant = -24.
- A. 2, 12
 - B. 2, -12
 - C. -2, 12
 - D. -2, -12

Scope IV: Data Types

16. _____ specifies the type of value to be stored and the operations that can be performed on it?
- A. Data Base
 - B. Data Source
 - C. Data Structure
 - D. Data Type
17. Vanilla-JavaScript is said to be _____ typed by default?
- A. Hybrid
 - B. Latent
 - C. Strongly
 - D. Weakly

Scope V: Data Structures

18. Which of the following Data Structures can implement both FIFO and LIFO operations?
- A. Array
 - B. Hash
 - C. Stack
 - D. Queue

19. Arrays are examples of _____ data types?
- A. Abstract
 - B. Composite
 - C. Enumerated
 - D. Primitive
20. Which is the correct syntax to declare and initialize an array?
- A. `var colors = array("red", "green", "blue")`
 - B. `var colors = ("red", "green", "blue")`
 - C. `var colors = ["red", "green", "blue"]`
 - D. `var colors = {"red", "green", "blue"}`
21. A *Pyramid Scheme* network model can be implemented using which of the following data structures?
- A. Array
 - B. Graph
 - C. Linked List
 - D. Tree

Scope VI: Control Structures 1 and 2

22. Decision Statements are derivatives of _____ constructs?
- A. Iteration
 - B. Selection
 - C. Sequence
 - D. Ternary
23. Which of the following Iteration construct is best suited for traversal?
- A. `do while`
 - B. `for`
 - C. `for-in`
 - D. `while`
24. Write a JavaScript program to compute the Sum of Multiples of 5 between 1 and 100.
- A. 1000
 - B. 1005
 - C. 1050
 - D. 1500
25. Write a JavaScript program to compute the 12th term of the *Fibonacci* sequence.
- A. 55
 - B. 89
 - C. 98
 - D. 144

26. Write a JavaScript program to convert 1759 to Hexadecimal.

- A. 10100000
- B. 10101000
- C. 10101100
- D. 10101110

Scope VII: Subprograms

27. Which of the following terms is not part of a Function definition?

- A. Function Body
- B. Function Call
- C. Function Header
- D. Function Parameters

28. A function that calls itself is called a _____ function?

- A. Arrow
- B. Lambda
- C. Nested
- D. Recursive

Scope VIII: Encapsulation

29. In Object-Oriented Programming, the method that initializes the properties of a class is known as?

- A. Constructor Method
- B. Display Method
- C. Getter Method
- D. Setter Method

30. Class properties should only be accessed through *setter* and *getter* methods.

- A. True
- B. False

31. Which Access Specifier denotes that only a class and its subclasses have access to its members?

- A. Private
- B. Protected
- C. Public
- D. Restrict

32. A Static Method cannot be invoked without the object of the class.

- A. True
- B. False

Scope IX: JavaScript built-in APIs

33. The following *wrapper* methods are used to format a string except?
- A. `bold()`
 - B. `italics()`
 - C. `strike()`
 - D. `underline()`
34. The _____ method is used to remove *whitespaces* from a string?
- A. `crop()`
 - B. `strip()`
 - C. `trim()`
 - D. `clear()`
35. The following methods can be used to extract characters from a string except?
- A. `slice()`
 - B. `split()`
 - C. `strip()`
 - D. `substr()`
36. The following methods can be used to format numbers except?
- A. `toExponential()`
 - B. `toFinite()`
 - C. `toFixed()`
 - D. `toPrecision()`
37. The _____ method is used to convert a string to number?
- A. `int()`
 - B. `parseInt()`
 - C. `toInt()`
 - D. `toNumber()`
38. The _____ method is used to convert a number to string?
- A. `str()`
 - B. `parseString()`
 - C. `toStr()`
 - D. `toString()`
39. _____ is used to implement x raised-to power y ?
- A. `Math.exp(x,y)`
 - B. `Math.pow(x,y)`
 - C. `power(x,y)`
 - D. `x.toPower(y)`

40. _____ is used to round-up a number?
- A. Math.round(x)
 - B. Math.toFixed(x)
 - C. round(x)
 - D. x.round()
41. _____ returns a random number?
- A. getRandom()
 - B. Math.rand()
 - C. Math.random()
 - D. random()
42. The following are Array methods except?
- A. index()
 - B. reduce()
 - C. splice()
 - D. unshift()
43. Which of the following is not an Array method?
- A. forEach()
 - B. indexOf()
 - C. orderBy()
 - D. valueOf()
44. Which of the following keywords can be used to access all values passed into a function call?
- A. args
 - B. arguments
 - C. parameters
 - D. this
45. Which of the following methods is not part of the JavaScript Date object?
- A. getDate()
 - B. getDay()
 - C. getMonth()
 - D. getYear()
46. In the month of August, `date.getMonth()` returns _____?
- A. 7
 - B. 8
 - C. Aug
 - D. August

Scope X: Regular Expression

47. All of these are application areas of Regular Expression except?
- A. Character Encoding
 - B. Data Validation
 - C. Pattern Matching
 - D. Search Optimization
48. Which of the following is not a valid regex method?
- A. exec()
 - B. match()
 - C. pattern()
 - D. test()

Scope XI: Errors and Debugging

49. The following are types of Program Errors except?
- A. Logical Errors
 - B. Semantic Errors
 - C. Syntax Errors
 - D. Typographic Errors
50. Logical errors are detected at what stage of program execution?
- A. Compile-time
 - B. Run-time
 - C. All of the above
 - D. None of the above

Scope XII: Exception Handling

51. _____ defines a block of code to be executed regardless of the Exception result?
- A. Catch
 - B. Finally
 - C. Throw
 - D. Try
52. The following are types of Exceptions except?
- A. ArithmeticException
 - B. NumberFormatException
 - C. StackOverflowException
 - D. FileNotFoundException

Scope XIII: Advanced Data Structures

53. Objects in JavaScript Object Notation (JSON) are written in which format?
- A. key: value
 - B. key: "value"
 - C. "key": value
 - D. "key": "value"
54. Which of the following commands is used to convert a JSON object to JavaScript object?
- A. JSON.parse()
 - B. JSON.toString()
 - C. parse()
 - D. toJSON()

Scope XIV: Multi-threading

55. Asynchronous JavaScript and XML (AJAX) is best suited for which of the following operations?
- A. Sending data to server
 - B. Reading data from server
 - C. Updating webpage content
 - D. All of the above

Scope XV: Server-side Scripting

56. Node.js was developed by _____ in the year _____?
- A. Anders Hejlsberg, 2012
 - B. Brendan Eich, 1995
 - C. Jeremy Ashkenas, 2011
 - D. Ryan Dahl, 2009
57. Which of the following is an application area of Node.js?
- A. Database Applications
 - B. Desktop Applications
 - C. Hardware Applications
 - D. All of the above
58. Node.js files are saved with _____ file extension?
- A. .js
 - B. .json
 - C. .node
 - D. .nodejs

59. In Node.js, which of the following methods is used to import the file system Module?

- A. `import('fs')`
- B. `include('fs')`
- C. `request('fs')`
- D. `require('fs')`

60. In Node.js, which of the following methods is used to retrieve the contents of a directory?

- A. `bufferFile()`
- B. `parseFile()`
- C. `readFile()`
- D. `streamFile()`